

sol1_enhancer_deep_agents

Take-home. LangChain Deep Agents ≥ 0.7 .

Python owns the loop. The doer **holds a write tool**, scoped to `tickets/**`. The judge still has none.

Model: `anthropic:claude-sonnet-5`.

What you will build

FILE	JOB
enhancer.py	poll-and-act in Python
roles.py	create_deep_agent, scoped write tool
skills/doer/SKILL.md	doer prompt
skills/judge/SKILL.md	judge prompt
check_stop.py	done, cost, max turns. Signature is not an exit
tests/	pytest, no SDK

Turn off the default `general-purpose` subagent. It ships harness FS tools. Leaving it on is how a "scoped" agent writes `app/`.

Learning objectives

- Hand each subagent its own tool list
- Put the path check **inside** the write tool
- Refuse with a sentence, not an exception
- Hide orchestrator write tools
- Compare `check_stop.py` to the plugin (signature is not an exit here)

Starting architecture

```
python3 loop.py --once --repo TARGET
└─ create_deep_agent
    orchestrator: no write_file, edit_file, delete, execute
    doer: write tool, tickets/** only
    judge: read_file only
    FilesystemBackend(root_dir=crm, virtual_mode=True)
```

Scope. Three ways, all used

1. Per-subagent tool list. Judge is never handed write.
2. Path check inside the doer's write tool.
3. Harness fence: virtual FS root, permissions deny on orchestrator, allow `tickets/**` on doer, deny on judge.

```
ORCHESTRATOR_EXCLUDED_TOOLS = {write_file, edit_file, delete, execute}
```

Refusal is a sentence

```
@tool(f"write_{role.name}")
def write(path: str, content: str) -> str:
    try:
        scope.check(path)
    except ScopeViolation:
        return f"REFUSED. {role.name} may write {allowed}. {path} is outside that scope."
    (repo / path).write_text(content, encoding="utf-8")
    return f"wrote {path}"
```

An unformatted exception in an agent's context tends to start a retry loop. A short sentence tends to change the next action.

`check_stop.py` is a different contract

```
def check(*, done, turns, max_turns, spent_usd=0.0, max_usd=2.0) -> dict:
    if done: return {stop: True, reason: "done"}
    if spent_usd >= max_usd: return {stop: True, reason: "cost"}
    if turns + 1 >= max_turns: return {stop: True, reason: "max turns"}
    return {stop: False, reason: None}
```

Repeated signature is **not** an exit. Stuck work burns turns or dollars until cost or max-turns fires. `MAX_USD = 2.0`. Done waits for human LGTM.

Marker is implemented

`MARKER = "<!-- enhancer-loop -->"`. Fetch `per_page=100` because the default 30 plus a marker filter wedges long threads.

`already_acted_on` compares GitHub ids as **ints**. `"1000000001" <= "999999999"` is True as text and would skip a real new comment.

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Commands

```
cd solutions/sol1_enhancer_deep_agents
python3 loop.py --table-only
task test && task table && task checks
task clone && task create-test-tickets
task run
```

Taskfile dotenv: ['../.env', '.env']. First file that defines a var wins.

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Expected table

role	writes	scope
orchestrator	no	nothing
doer	yes	tickets/**
judge	no	nothing

If judge prints `yes`, stop.

Troubleshooting

SYMPTOM	CAUSE	FIX
Writes app/	default general-purpose still on	turn it off
Retry storm	write tool raises	return a sentence
Skipped a new comment	id compared as text	compare as ints
Stopped on same gaps	copied plugin check_stop	signature is not an exit here

Validation

- `[] table: judge no`
- `[] pytest green without a key`
- `[] doer write to tickets/T900.md works`
- `[] doer write to app/main.py returns REFUSED`
- `[] marker on posted comments`

GitHub Actions

ENHANCER_BACKEND=deep-agents. **Secret** ANTHROPIC_API_KEY.

```
python3 loop.py --once --repo "$GITHUB_WORKSPACE" --ticket "$TICKET"
```

Same copy-me workflow as the other sol ports.

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Recap

Deep Agents is a harness. The subagent tools list replaces the parent. Python still owns the outer budget.

Three belts: missing tool, in-tool path check, harness profile. All three.

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